Guobin Zhao

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Education

Pusan National University — Busan, KR
Ph.D. in Chemical Engineering, GPA: 3.96/4.50 — 2020–2025

- Thesis: "Accelerating Nanoporous Materials Discovery for Adsorptive Separations: from Database to Machine Learning Models"

Jimei University — Xiamen, CN
M.S. in Naval Architecture and Ocean Engineering — 2018–2020

- Thesis: "Study on Adsorption Storage of Methane on Typical MOFs"

Zhejiang Ocean University — Zhoushan, CN
B.S. in Marine Engineering (Ship Building and Repairing) — 2014–2018

- Thesis: "Design of a Stacked-Type Hatch Cover"

Experience

National University of Singapore — Singapore, SG
Postdoctoral Research Fellow at MACE Group — 2025.12-2026.12

- Fine-tuning LLM for Validation of MOFs
- Validate if LLM understand crystal structures, and give the reason for unphysical structures.
- Graph Transformer - Based Method for Solvent Removal of MOFs
- Develop a machine learning model for free & coordinated solvent removal for MOFs.
- Refinement of MOF Structures by Generative AI
- Explore the application of generative AI in refining MOF structures.
- CO_2 Conversion by MOF
- Explore performance of CO_2 conversion by CALF-20 by DFT simulation.

Pusan National University — Busan, KR
Postdoctoral Associate at Chung Research Group — 2025.09-2025.11

- CoRE MOF 2026 Database (Co-work with Prof. Kulik Group at Massachusetts Institute of Technology and Prof. Moosavi Group at University of Toronto.)
- Updated the CoRE MOF database and investigated the sensitivity of not computation-ready structures on machine learning models.
- Machine Learning Potential for MOFs (Co-work with Prof. Rosen Group at Princeton University.)
- High-throughput density functional theory calculations and training machine learning force fields.
- Zn-MOF for Direct Air Capture (Co-work with Prof. Koh Group at KAIST.)
- Found stable structure with high performance for carbon capture.

- Quantum Machine Learning Discovery of Nanoporous Materials for Direct Air Carbon Capture (Co-work with Prof. Moosavi Group at University of Toronto.)
- Analyzed the structures that simulation fail.

Publications

[1] **G. Zhao**, Y. Chen, A. K. Ball, C. Oh, J. Huang, S. M. Moosavi, H. J. Kulik, and Y. G. Chung, "High-openness updates of the core mof database and sensitivity to ncr structures in machine learning", *Ready for submission*, 2026.

[2] **G. Zhao** and X. Y. Li, "Exploring large language models for metal-organic framework validation", *Submitted*, 2026.

[3] J. Guo, X. Xu, L. E, **G. Zhao**, and Y. Li, "Beyond disappearance: Heterojunction aops and the hidden carbon fate in water systems", *ACS ES&T Water*, 2026.

[4] Y. Chen, **G. Zhao**, L.-C. Lin, and Y. G. Chung, "A Master Isotherm Model Approach to Quantify Defects in UiO-66 from Nitrogen Adsorption Isotherms", *Langmuir*, vol. 41, no. 44, pp. 29 851–29 858, 2025.

[5] **G. Zhao**, L. M. Brabson, S. Chheda, J. Huang, H. Kim, K. Liu, K. Mochida, T. D. Pham, G. G. Terrones, S. Yoon, et al., "CoRE MOF DB: A Curated Experimental Metal–Organic Framework Database with Machine–Learned Properties for Integrated Material–Process Screening", *Matter*, vol. 8, no. 6, 2025.

[6] **G. Zhao**, P. Zhao, and Y. G. Chung, "MOFClassifier: A Machine Learning Approach for Validating Computation-Ready Metal–Organic Frameworks", *Journal of the American Chemical Society*, vol. 147, no. 37, pp. 33 343–33 349, 2025.

[7] S. Yoon, J. Jung, **G. Zhao**, Y. Lee, K. B. Lee, and Y. G. Chung, "Techno-Economic Analysis of Cu (I)–Loaded Porous Carbons for CO Separation via Pressure/Vacuum Swing Adsorption Process", *Separation and Purification Technology*, vol. 378, no. 3, p. 134 786, 2025.

[8] Y. Chen, **G. Zhao**, S. Yoon, P. Habibi, C. S. Hong, S. Li, O. A. Moultos, P. Dey, T. J. Vlugt, and Y. G. Chung, "Computational Exploration of Adsorption–Based Hydrogen Storage in Mg-Alkoxide Functionalized Covalent–Organic Frameworks (COFs): Force–Field and Machine Learning Models", *ACS applied materials & interfaces*, vol. 16, no. 45, pp. 61 995–62 009, 2024.

[9] Y. C. Chung, C. W. Yang, and **G. Zhao**, "Global Warming Potential (GWP) Prediction Apparatus based on Computational Chemistry and Machine Learning", 10-2023-0061292, Patent, 2024.

[10] **G. Zhao** and Y. G. Chung, "PACMAN: a Robust Partial Atomic Charge Predicter for Nanoporous Materials Based on Crystal Graph Convolution Networks", *Journal of Chemical Theory and Computation*, vol. 20, no. 12, pp. 5368–5380, 2024.

[11] **G. Zhao**, H. Kim, C. Yang, and Y. G. Chung, "Leveraging Machine Learning to Predict the Atmospheric Lifetime and the Global Warming Potential of SF_6 Replacement Gases", *The Journal of Physical Chemistry A*, vol. 128, no. 12, pp. 2399–2408, 2024.

[12] **G. Zhao**, Y. Chen, and Y. G. Chung, "High-Throughput, Multiscale Computational Screening of Metal–Organic Frameworks for Xe/Kr Separation with Machine–Learned Parameters", *Industrial & Engineering Chemistry Research*, vol. 62, no. 37, pp. 15 176–15 189, 2023.

[13] S. Han, S. Lee, D. Kim, J. Seong, A. Sharma, J. Lim, S. Jeong, **G. Zhao**, Y. Chen, S. B. Baek, et al., "Construction of Chimeric Metal–Organic Frameworks with Symmetry-Mismatched Building Blocks", *Chemistry of Materials*, vol. 35, no. 15, pp. 5903–5913, 2023.

[14] **G. Zhao**, Q. Zheng, X. Zhang, and W. Zhang, "Adsorption Equilibrium and the Effect of Honeycomb Heat Exchanging Device on Charge/Discharge Characteristic of Methane on MIL-101 (Cr) and Activated Carbon", *Chinese Journal of Chemical Engineering*, vol. 28, no. 7, pp. 1964–1972, 2020.

[15] X. Zhang, Q. Zheng, **G. Zhao**, and W. Zhang, "Adsorption Equilibrium and Charge/Discharge Characteristics of Hydrogen on MOFs", *Cryogenics*, vol. 112, p. 103 121, 2020.

[16] X. Zhang, Q. Zheng, W. Zhang, and **G. Zhao**, "Study of Adsorption of Ship's Bilge Oily Water on Expanded Graphite/Activated Carbon Composite Adsorbents", *Ship Science and Technology*, vol. 42, no. 6, pp. 105–109, 2020.

[17] **G. Zhao**, Q. Zheng, W. Zhang, and X. Zhang, "Adsorption Equilibrium and Charge/Discharge Characteristics of Methane on MIL-101", *Journal of Fuel Chemistry and Technology*, vol. 47, pp. 1529–1936, 2019.

DEVELOPMENT

- **CoRE MOF** Computation-Ready Experimental Metal-Organic Framework — database & package
  PyPi • Github • Website • Zenodo
- **PACMAN-charge** Partial Atomic Charges for Porous Materials based on GCN — package
  PyPi • Github • Website
- **MOFClassifier** A Machine Learning Approach for Validating Computation-Ready MOFs — package
  PyPi • Github
- **ToBaCCo-MOF** Topologically Based Crystal Constructor — package (a refactored version)
  PyPi
- **SESAMI** Characterization Tools for Porous Materials Using N_2/Ar Adsorption — package (a refactored version)
  PyPi

Teaching

- **Teaching Assistant** at Pusan National University — Spring 2025
  Introduction to Computational Nano Science and Engineering (CG3500768-062)
- **Teaching Assistant** at Pusan National University — Fall 2024
  Current Events in Chemical and Biomolecular Engineering (CG1501061-062)
- **Teaching Assistant** at Pusan National University — Spring 2024
  AI for Chemical and Materials (CG2001538-062)
- **Teaching Assistant** at Pusan National University — Fall 2022
  Introduction to Data Science for Chemical Engineers (CG3500748-062)

Skills

- **Machine Learning Tools:** PyTorch, Scikit-learn, XGBoost, Pymatgen, ASE, RDKit, MLPs (MACE, CHGNET, SevenNet)
- **Density Functional Theory:** VASP, CP2K, Gaussian
- **Molecular Simulation:** Lammps, RASPA, Materials Studio
- **Visualization:** VESTA, Ovito, Avogadro, Matplotlib, Seaborn
- **Programming Languages:** Python, MATLAB, Shell, Markdown
- **Languages:** Chinese (Native), English (Professional Knowledge), Korean (Basic Knowledge)

Research Interests

- **AI4Science:** Accerlerating materials discovery using artificial intelligence techniques.
- **Big Data:** Using natural language processing and large language model to construct MOF database.
- **High-Throughput Screening:** Multi-scale screening for gas storage and separation.

Scholarships and Awards

- CBE FOCUS (JACS work) in School of Chemical Engineering at Pusan National University 2025
- CBE FOCUS (Matter work) in School of Chemical Engineering at Pusan National University 2025
- PNU Research (Matter work) at Pusan National University 2025
- Featured in the JCTC editorial on machine learning and statistical mechanics for contributing to next-generation chemical theory integration (JCTC work, Cover) 2025
- Excellent Poster Award (KSIEC-Busan) 2024
- Excellent Poster Award (KIChE-Busan) 2024
- Published as part of the Industrial & Engineering Chemistry Research (IECR work) virtual special issue "2023 Class of Influential Researchers" 2023
- Excellent Poster Award (KIChE-Busan) 2023
- Excellent Poster Award (KSIEC-Busan) 2023
- Excellent Poster Award (KSIEC-Busan) 2021
- Foreign Student Scholarship 2020–2021
- Excellent Professional Master's Degree Thesis Award in Fujian Province 2020
- Excellent Master's Thesis Award of Jimei University 2020
- Excellent Master's Degree Graduate 2020
- M.S. Scholarship 2018–2019
- B.S. Scholarship 2017–2018
- Excellent Undergraduate Thesis Award 2018

Extracurricular Activities

- Peer review for ACS Applied Nano Materials, JACS Au, ACS Omega 2024-2026

- Poster presentation (South Korea) 2021-2024
  The Korean Society of Clean Technology, The Korean Institute of Chemical Engineers, Korean Chemical Society, The Korean Society of Industrial and Engineering Chemistry
- Poster presentation (International) 2024
  9th International Conference on Metal-Organic Frameworks and Open Framework Compounds (MOF2024 at Singapore)
- Oral presentation 2024
  - Korean Chemical Society, "PACMAN: A Robust Partial Atomic Charge Predicter for Nanoporous Materials Based on Crystal Graph Convolution Networks"
  - The Korean Society of Clean Technology, "High-Throughput, Multiscale Computational Screening of Metal-Organic Frameworks for Xe/Kr Separation"

International Collaborators

- North America:
  - *Snurr Group @ Northwestern*
  - *HJK Group @ MIT*
  - *Rosen Research Group @ Princeton*
  - *Siepmann Group @ Minnesota*
  - *Sholl Group @ Georgia Tech*
  - *Moosavi Group @ Toronto*
- Europe:
  - *FX Coudert Group @ PSL University*
  - *Haranczyk @ IMDEA*
  - *Thijs Group @ TU Delft*
- Asia:
  - *Song Li @ HUST*
  - *Koh @ KAIST*
  - *Lah Group @ UNIST*
  - *Hong Group @ KU*
  - *Lin Group @ NTU & OSU*